

Zenghui Sun

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EDUCATION

Georgia Institute of Technology

Atlanta, GA • Aug 2024 – May 2026

M.S. Computer Science, Machine Learning

GPA: 4.0 / 4.0

Coursework: Graduate Algorithms, Deep Learning, Deep Reinforcement Learning, Computer Vision, Computational Data Analysis, Data & Visual Analytics

Northeastern University

Boston, MA • Dec 2023

B.S. Computer Science, Artificial Intelligence (Summa Cum Laude)

GPA: 3.93 / 4.0

Coursework: Algorithms & Data, Computer Systems, Networks & Distributed Systems, Object-Oriented Design, Theory of Computation, Software Engineering

Teaching Assistant, CS 2510 Fundamentals of Computer Science 2 (object-oriented programming and data structures in Java)

WORK EXPERIENCE

Pango Valve System, AI-LLM Application Engineer

Remote (Atlanta, GA) • Jul 2026 – Present

- As technical owner, architected a production enterprise retrieval system from 0 to 1: designed a three-store backend (vector/full-text search index, structured SQL parameter database, object store) and a dual-path request router (rules-first regex plus LLM classifier) sending exact-parameter lookups to SQL and open questions to hybrid vector + BM25 retrieval with reranking and mandatory citations
- Designed the routing and quoting layer as a Python FastAPI service with a swappable retrieval engine, specified a multimodal ingestion pipeline (OCR and table-structure parsing, Whisper ASR for video, CAD title-block/BOM extraction), and benchmarked on-prem serving on NVIDIA DGX Spark (GB10, ARM, 128GB) against hosted APIs

Georgia Institute of Technology, Research Assistant

Atlanta, GA • Jun 2025 – Present

Advisors: Dr. Kai Wang & Dr. Gian-Gabriel Garcia · Collaborator: Children's Healthcare of Atlanta (CHOA)

- Built 9-stage preprocessing pipeline in Python for irregularly-sampled PICU EHR flowsheets: WDL-tree-aware semantic imputation, LLM chain-of-thought severity scoring (0–1 continuous) replacing ordinal encoding, and 4 temporal delta features (admission-relative, inter-measurement, historical-extrema, age-normalized deviations)
- Defined model API contract and payload specification for Epic FHIR integration within a cross-institutional ~19-member team (CHOA + GT) using PR-based development with pre-commit hooks
- Built and benchmarked a suite of models for pediatric ICU-transfer prediction (reconstruction-augmented LSTM, MIST transformer, neural network, decision tree, isolation forest, and logistic regression with a bagged ensemble) against the clinical industry-standard PEWS, raising AUROC from 0.819 (PEWS) to 0.911, with the MIST transformer best overall
- Deployed the bagged logistic-regression model into CHOA production clinical workflow via Epic EHR API, chosen over the higher-performing MIST transformer for interpretability and lower deployment complexity (AUROC 0.911, Recall 0.742 at 0.617% event rate, 3.5M-row test set)
- Presented feature-importance and time-window analyses to CHOA clinical staff; iterated 2-hour and 12-hour prediction windows and alert thresholds based on direct physician feedback

Winning Health, Algorithm Engineer Intern

Shanghai, China • Jul 2024 – Aug 2024

- Built end-to-end post-training pipeline for Llama 3 8B-based medical LLM: designed schema-constrained instruction templates for clinical entity extraction, generated 2,000+ QA/preference pairs via LLM ensemble, and trained reward model scoring factuality, completeness, and JSON adherence
- Achieved 31.7% improvement on domain-specific extraction benchmark over existing baselines; integrated trained model into WinGPT production corpus for clinical information-extraction deployment

PROJECTS

ICU Patient Allocation via Restless Multi-Armed Bandits (eICU)

github.com/Eric-szh/RMAB-MIMIC • 2025

- Formulated ICU bed allocation as finite-horizon RMAB with 82-state MDP; estimated transition matrices from 17,885 eICU patient-day transitions and derived a Q-function heuristic policy (value iteration, $\gamma=0.97$) that reduced simulated deaths by ~39 vs. sickest-first baseline over a 90-day horizon (N=300, 20 seeds)

RGB-LiDAR Fusion for Food Macro-Nutrient Estimation (Nutrition5k)

2026

- Designed a dual-stream PyTorch model (pretrained ResNet18 for RGB, purpose-built 4-layer CNN for depth) with a fused regression head, cutting calorie MAE 17.6% (57.70 to 47.52 kcal) over an RGB-only baseline; ablations showed a lightweight depth encoder beats a full ResNet18 depth branch

RESEARCH OUTPUT

- Shangqing Xu, **Zenghui Sun**, Soumil Sahu, Kai Wang, B. Aditya Prakash. "Multimodal Irregular Time-series Analysis on Clinical Databases." Manuscript under review, NeurIPS 2026.
- Hanmo Zhang, **Zenghui Sun**, Kai Wang. "Networked Restless Multi-Arm Bandits with Reinforcement Learning." arXiv:2512.06274 [cs.LG], 2025.
- "Early Prediction of Pediatric ICU Transfer: A Comparative Study of Tabular and Time Series Machine Learning Models." Manuscript in preparation, to be submitted to the INFORMS Journal on Data Science.

SKILLS

- Languages: Python, Java, JavaScript / Node.js, SQL
- Core CS / Software: Data structures and algorithms, object-oriented design, distributed systems, REST APIs, FastAPI, Express, Git, Jupyter
- ML / AI: PyTorch, scikit-learn, LLM fine-tuning, reward modeling, RLHF, LSTM, CNN, RAG, hybrid retrieval (vector + BM25), reranking, embeddings